



PRODUCT NAME

Professional Organic Shower Gel Formula

Description

This professional organic shower gel provides gentle cleansing with a pH-balanced formula. It is biodegradable and free from synthetic fragrances, making it an eco-friendly yet effective cleansing solution.

Technical Specifications

pH Level:	7
Viscosity:	Medium
Shelf Life:	18 months
Batch Size:	10-100 kg
Processing Time:	2 hours
Temperature:	Ambient
Storage Conditions:	Store in a cool, dry place away from direct sunlight.
Equipment:	Mixing tank, stirrer, pH meter
Certification:	Certified Organic Ingredients

Formulation Table

Sr.No	Ingredient	INCI Name	%	Function
1	Aqua	Aqua	65.0%	Solvent
2	Coco-Glucoside	Coco-Glucoside	10.0%	Surfactant
3	Sodium Coco-Sulfate	Sodium Coco-Sulfate	8.0%	Cleansing agent
4	Glycerin	Glycerin	5.0%	Humectant
5	Xanthan Gum	Xanthan Gum	0.5%	Thickener
6	Citric Acid	Citric Acid	0.3%	pH Adjuster
7	Aloe Barbadensis Leaf Juice Powder	Aloe Barbadensis Leaf Juice	0.5%	Skin conditioning agent
8	Sodium Benzoate	Sodium Benzoate	0.2%	Preservative
9	Potassium Sorbate	Potassium Sorbate	0.2%	Preservative
10	Natural Fragrance	Fragrance (Natural)	0.3%	Adds a natural scent without synthetic aroma
Total			100%	

Manufacturing Process

Phase 1: Phase A

1. Mix Aqua and Aloe Barbadensis Leaf Juice Powder in a mixing tank until fully dissolved.
2. Add Glycerin to the mixture and stir until homogenous.

Phase 2: Phase B

1. In a separate vessel, combine Coco-Glucoside and Sodium Coco-Sulfate.
2. Add Phase A to Phase B slowly while stirring continuously.

Phase 3: Phase C

1. Add Xanthan Gum to the mixture while stirring to ensure uniform thickening.
2. Adjust pH to 7 using Citric Acid solution.

Phase 4: Phase D

1. Incorporate Sodium Benzoate and Potassium Sorbate into the formulation as preservatives.
2. Add Natural Fragrance last and mix until evenly distributed.

Required Equipment

Mixing tank, stirrer, pH meter

Safety Precautions

Handling: Wear appropriate PPE including gloves, safety glasses, lab coat.

PPE Requirements: Chemical-resistant gloves, safety goggles, protective clothing.

Storage: Store in cool, dry place away from direct sunlight. Keep containers tightly closed.

Storage Conditions: Store in a cool, dry place away from direct sunlight.

Packaging Notes

Packaging: Use chemically compatible containers (HDPE, glass, or PET).

Labeling: Include product name, ingredients, usage instructions, and safety warnings.

Certification: Certified Organic Ingredients

Scaling Note

Lab Scale: This formulation is designed for laboratory testing and development.

Pilot Scale: For pilot production, scale proportionally and verify all parameters.

Production Scale: Consider equipment limitations, mixing efficiency, and process validation.

Batch Size: Current formulation is optimized for 10-100 kg.

Scaling Factor: Maintain ingredient ratios while adjusting processing parameters as needed.

Product Usage Instructions & Application Guidelines

Apply a small amount of the gel onto a wet sponge or directly onto skin. Work into a lather and cleanse the entire body. Rinse thoroughly with water.